

ALGORITHMIC FORENSICS APPENDIX

How I use AI. I used Claude Code, Anthropic's coding agent, as a research assistant and programmer. It did three jobs: - **Search:** separate AI sub-agents searched the web for flood events, media coverage, relocation sites and governance evidence. - **Code:** it wrote PowerShell scripts that turn that evidence into datasets. - **Build:** it built the maps and charts.

I set the questions, the framing and the thesis. Every number in the project comes from a script in `methods/scripts/`, not from AI prose.

Workflow and prompts. Each research prompt, logged in `methods/agent_prompts.md`, carried the same hard rules: - record verbatim quotes with URLs; - never invent a number, quote or link; - give every claim a confidence rating (high, medium or low); - mark gaps as "not_found".

Agents wrote structured JSON to `data/raw/`. Scripts then merge, geocode and score those files into the published datasets. External data come from official sources: Census 2011 language by place, JRC Global Surface Water (Landsat, 1984–2021), eThekweni's 1-in-100-year floodplain, and WMO station records.

Checking for error and bias. I checked in four ways: 1. **Figures against sources.** Every figure in the write-ups was checked against its source file by text search. 2. **Second agents.** Separate agents re-verified the claims of earlier ones. 3. **Baselines.** Each spatial claim was compared with random locations. 4. **Counterexamples.** I asked for cases where my thesis fails: relocations that worked, and places where warnings in the local language still failed.

To limit bias: - **Language:** analysed at the place level only, never by inferring any individual's ethnicity. - **Victims:** not named. - **Case selection:** I label the international comparison as outcome-selected, meaning a pattern rather than a test.

One example where I caught the AI. An early dataset flagged six relocation sites as flooded after people moved in. A second verification agent re-checked every source and three flags failed: - **Sondela:** the "flood-prone wetland" described the old settlement, not the new site. - **Delft TRA 5:** no source named the site. - **Lamontville:** the February 2025 deaths were at the Gwala Street camp, not the camp the list called "Barcelona 2".

I corrected the data (`methods/scripts/apply_reflood_sweep.ps1`). I also fixed the AI's constitutional citations: "distinctive", not "distinct"; Schedule 4B, not Schedule 5; Disaster Management Act s26/40/54, not s41/55.

How the visual metrics are calculated. - **Floodplain exposure:** each Durban site is queried against the city's floodplain as inside, within 250 m, or within 1 km. The share is the count divided by 28 sites. - **Baseline:** 300 random points, generated with a fixed seed (20260927) so the same points come out on every run, across Durban's urban land, run through the same test. - **Satellite water:** the share of 30 m Landsat pixels within 1 km that were wet less than 50% of the time. - **Factor chart:** 11 cases coded yes, partial or no on 10 factors. The share

is (yes + ½ partial) ÷ the number of cases with evidence. - **Years "temporary"**: 2026 minus the year the site was established.

Is it statistically defensible? The visuals are descriptive, and I present them that way.

The key spatial claim changed because of a test. I expected the camps to cluster in the floodplain. They don't:

| Measure | Camps | Random ground | |---|---|---| | Inside or within 250 m | 39% | 34% | |
Chance of 11 or more of 28 if camps were placed at random (binomial) | $p = 0.34$ | |

So the text says the flood map "made no difference to siting", not that sites were targeted.

Known limits, stated on the charts: - Site coordinates are accurate only to $\pm 1-3$ km. - Reflood counts are minimums: only 30 of 84 sites were searched individually. - The 11-case comparison is small and outcome-selected.

The whole pipeline can be rerun from REPRODUCE.md.

(About 490 words)